

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Questions

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks 50

Duration :60 Minutes

Annexure A

Scenario-Based Questions

Code of Conduct:

*I **PAVAN JANA (192472313)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: **PAVANJ**

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

Scenario-Based Questions

1. Object Boundary Detection

In a medical image, boundaries between tissues need to be clearly identified.

- (a) Interpret the scenario and explain the importance of boundary detection.
- (b) Identify key challenges in detecting boundaries.
- (c) Apply a suitable edge detection method.
- (d) Justify your selection with logical reasoning.
- (e) Present your explanation clearly and systematically.

Object Boundary Detection

(a) Interpret the scenario and explain the importance of boundary detection

- **Medical images contain different tissues and organs:** Images such as MRI, CT, and X-ray scans show various body structures that need to be clearly distinguished.
- **Boundary detection identifies tissue edges:** It detects the borders between different tissues, making each structure easier to recognize.
- **Helps detect abnormalities:** Clear boundaries allow doctors to identify tumors, fractures, or damaged tissues more accurately.
- **Improves diagnosis:** Accurate boundary detection provides reliable information for medical diagnosis and treatment planning.
- **Supports image analysis:** It is an important step in medical image segmentation and computer-aided diagnosis.

(b) Identify key challenges in detecting boundaries

- **Image noise:** Random noise creates unwanted edges, making boundary detection less accurate.
- **Low contrast:** Similar intensity values between tissues make it difficult to distinguish their boundaries.
- **Blurred edges:** Some tissue boundaries are not sharp, making edge detection more challenging.
- **Overlapping structures:** Different tissues may overlap, causing confusion in identifying the correct boundary.
- **Variations in image quality:** Differences in brightness, resolution, or scanning conditions can affect detection accuracy.

(c) Apply a suitable edge detection method

Method: Canny Edge Detection

- **Noise reduction:** A Gaussian filter is applied to remove unwanted noise from the image.
- **Gradient calculation:** The algorithm calculates changes in pixel intensity to locate possible edges.
- **Non-maximum suppression:** Weak and unnecessary edges are removed to keep only the strongest edges.
- **Double thresholding:** Strong and weak edges are identified using two threshold values.
- **Edge tracking by hysteresis:** Connected weak edges are preserved while isolated false edges are removed, producing accurate boundaries.

(d) Justify your selection with logical reasoning

- **High accuracy:** The Canny algorithm detects edges more precisely than many other edge detection methods.
- **Noise resistant:** It reduces the impact of noise before detecting boundaries.
- **Detects weak and strong edges:** This helps identify fine tissue boundaries that simpler methods may miss.
- **Reduces false edges:** The double thresholding and hysteresis process minimizes incorrect edge detection.
- **Suitable for medical images:** Its ability to produce clear and continuous boundaries makes it ideal for medical image analysis.

(e) Present your explanation clearly and systematically

- **Boundary detection** is essential for separating tissues and organs in medical images.
- **Challenges** such as noise, low contrast, and blurred edges make detection difficult.
- **Canny Edge Detection** effectively overcomes these challenges through noise removal and accurate edge identification.
- **The algorithm produces clear, continuous, and reliable boundaries**, making it highly suitable for medical imaging.
- **As a result, it improves diagnosis, image segmentation, and treatment planning**, leading to better healthcare outcomes.

2. Simple Segmentation

An image contains objects clearly distinguishable from the background based on intensity.

- (a) Interpret the scenario and explain segmentation requirements.
- (b) Identify key features enabling separation.
- (c) Apply a suitable segmentation technique.
- (d) Justify your decision logically.
- (e) Present your response clearly.

Simple Segmentation

(a) Interpret the scenario and explain segmentation requirements

- **Image contains distinct objects:** The objects have different intensity values compared to the background.
- **Goal of segmentation:** Separate the object region from the background accurately.
- **Clear object extraction:** The segmented image should preserve the complete object shape.
- **Support further processing:** Segmentation helps in object recognition, measurement, and analysis.

(b) Identify key features enabling separation

- **Intensity difference:** Objects and background have different pixel intensity values.
- **High contrast:** Clear contrast makes separation easier.
- **Uniform background:** A consistent background improves segmentation accuracy.
- **Well-defined boundaries:** Object edges are clearly visible.
- **Minimal noise:** Less noise results in better segmentation.

(c) Apply a suitable segmentation technique

Technique: Thresholding

- Select an appropriate threshold value.
- Compare each pixel intensity with the threshold.
- Pixels above the threshold are classified as the object.
- Pixels below the threshold are classified as the background.
- Generate a binary image with separated objects.

(d) Justify your decision logically

- **Simple implementation:** Easy to understand and apply.
- **Fast processing:** Requires less computation time.
- **Accurate for clear intensity differences:** Works well when object and background are easily distinguishable.
- **Efficient:** Produces reliable segmentation results with minimal complexity.
- **Widely used:** Suitable for many medical and industrial image-processing applications.

(e) Present your response clearly

- Thresholding is the most suitable segmentation method for images with clear intensity differences.
- It effectively separates objects from the background using a threshold value.
- The method is simple, fast, and accurate for this type of image.
- Proper segmentation improves object detection, image analysis, and decision-making in image processing.

3. Complex Segmentation

An image contains multiple regions with overlapping intensity values.

(a) Interpret the scenario and explain segmentation challenges.

(b) Identify key issues such as intensity overlap.

(c) Apply an appropriate advanced segmentation method.

(d) Justify how the method handles complexity.

(e) Provide a structured and clear explanation.

Complex Segmentation

(a) Interpret the scenario and explain segmentation challenges

- **Multiple image regions:** The image contains several objects or tissues with similar intensity values.
- **Complex segmentation:** Objects cannot be separated using simple thresholding.
- **Need for accurate separation:** The method should correctly identify each region despite intensity similarities.
- **Supports detailed analysis:** Accurate segmentation improves object recognition and medical diagnosis.

(b) Identify key issues such as intensity overlap

- **Intensity overlap:** Different regions have similar pixel intensity values.
- **Image noise:** Noise affects the accuracy of segmentation.
- **Blurred boundaries:** Object edges are not clearly visible.
- **Overlapping objects:** Adjacent regions make separation difficult.
- **Intensity variation:** Brightness changes within the same object reduce segmentation accuracy.

(c) Apply an appropriate advanced segmentation method

Technique: Region Growing

- Select one or more seed points in the image.
- Compare neighboring pixels with the seed.
- Add similar pixels to the region based on intensity.
- Continue growing until no matching pixels remain.
- Form separate regions representing different objects.

(d) Justify how the method handles complexity

- **Handles intensity overlap:** Groups pixels based on similarity rather than a single threshold.
- **Produces accurate regions:** Preserves object shapes and boundaries.
- **Reduces incorrect segmentation:** Expands only into similar neighboring pixels.
- **Suitable for medical images:** Effectively separates tissues with overlapping intensity values.
- **More reliable than thresholding:** Works well for complex images with multiple regions.

(e) Provide a structured and clear explanation

- Complex segmentation is required when objects have similar intensity values.
- Region Growing starts from seed points and groups similar neighboring pixels.
- It accurately separates overlapping regions while preserving object boundaries.
- This method improves image segmentation, analysis, and decision-making in medical imaging.

4. Broken Edges

After applying edge detection, object boundaries appear discontinuous.

- (a) Interpret the scenario and explain why edges are broken.
- (b) Identify key issues affecting boundary continuity.
- (c) Apply a suitable method to improve edge connectivity.
- (d) Justify your approach logically.
- (e) Present your explanation clearly.

Broken Edges

(a) Interpret the scenario and explain why edges are broken

- **Discontinuous boundaries:** The detected object edges are not connected completely.
- **Incomplete edge detection:** Some edge pixels are missing after edge detection.
- **Need for continuous boundaries:** Connected edges are important for accurate object detection and segmentation.
- **Affects image analysis:** Broken edges reduce the accuracy of object recognition.

(b) Identify key issues affecting boundary continuity

- **Image noise:** Noise causes gaps in detected edges.
- **Weak edges:** Low-intensity edges may not be detected.
- **Blurred boundaries:** Unclear object edges create discontinuities.
- **Improper threshold values:** Incorrect thresholds remove important edge pixels.
- **Low image quality:** Poor contrast affects edge continuity.

(c) Apply a suitable method to improve edge connectivity

Technique: Morphological Closing

- Apply **dilation** to connect nearby broken edges.
- Apply **erosion** to restore the original object shape.
- Fill small gaps between edge pixels.
- Produce smooth and continuous object boundaries.
- Improve the overall edge connectivity.

(d) Justify your approach logically

- **Connects broken edges:** Fills small gaps between disconnected boundaries.
- **Improves continuity:** Produces smooth and complete object outlines.
- **Reduces small holes:** Eliminates tiny gaps without changing the object significantly.
- **Enhances segmentation:** Makes object detection more accurate.
- **Suitable for image processing:** Commonly used after edge detection to improve results.

(e) Present your explanation clearly

- Broken edges occur due to noise, weak edges, and poor image quality.
- Morphological Closing connects nearby edge pixels and fills small gaps.
- It creates continuous object boundaries while preserving the object's shape.
- This improves edge detection, segmentation, and overall image analysis.

5. Combined Requirement

An industrial inspection system requires noise removal while preserving important edges for defect detection.

(a) Interpret the scenario and define the combined requirement.

(b) Identify key issues involving noise and edge preservation.

(c) Apply an appropriate sequence of techniques.

(d) Justify the order and selection with strong reasoning.

(e) Present your response in a clear and structured format.

Combined Requirement

(a) Interpret the scenario and define the combined requirement

- **Industrial inspection system:** The system is used to identify defects in manufactured products.
- **Need for noise removal:** Unwanted noise must be removed to improve image quality.
- **Need for edge preservation:** Important edges of defects must remain clear for accurate detection.
- **Combined goal:** The process should reduce noise while maintaining important object boundaries.

(b) Identify key issues involving noise and edge preservation

- **Image noise:** Random variations can create false defect patterns.

- **Loss of edges:** Excessive filtering may remove important defect boundaries.
- **Low contrast defects:** Small defects may be difficult to identify.
- **Blurred boundaries:** Noise reduction techniques can affect edge sharpness.
- **Trade-off problem:** Removing noise while preserving details is challenging.

(c) Apply an appropriate sequence of techniques

Technique: Gaussian Filtering + Canny Edge Detection

- **Step 1: Gaussian Filtering**
 - Removes unwanted noise from the image.
 - Smooths small intensity variations.
- **Step 2: Canny Edge Detection**
 - Detects important defect boundaries.
 - Preserves strong and weak edges accurately.
- **Step 3: Edge Analysis**
 - Uses detected edges to identify defects in the product.

(d) Justify the order and selection with strong reasoning

- **Noise removal first:** Filtering reduces unwanted variations before edge detection.
- **Prevents false edges:** Removing noise improves edge detection accuracy.
- **Canny after filtering:** Detects clear boundaries without being affected by noise.
- **Maintains important details:** The method preserves defect edges while improving image quality.
- **Suitable for inspection systems:** Provides accurate defect detection with reliable results.

(e) Present your response clearly and structured

- Industrial images require both noise reduction and edge preservation for accurate defect detection.
- Gaussian filtering is applied first to remove noise.
- Canny Edge Detection is then used to identify important edges.
- This sequence improves image quality while maintaining defect boundaries.
- The combined approach provides accurate and efficient industrial inspection results.